

DietHaven Limited

DietHaven Consult

Application Development Guide — Module 1

Prepared for: Software Developer

Tagline: "Your Food, Your Medicine"

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1. Purpose of This Document

This document describes the functional and technical requirements for Module 1 of the DietHaven Consult application. It is intended to guide the development team through the features, workflows, data structure, and clinical logic required to build a working platform for dietitians and patients.

DietHaven Consult digitizes the Nutrition Care Process (NCP) using the International Dietetics and Nutrition Terminology (IDNT) framework, giving dietitians a structured, standardized way to assess, diagnose, treat, and monitor patients, while embedding Nigerian-specific tools such as the Nigerian Food Exchange List and AI-assisted clinical support.

2. Brand Identity

All UI screens, buttons, charts, and printable reports should follow the DietHaven brand palette below:

Color Role	Description	Suggested Hex
Primary — Green	Main brand color. Navigation bars, primary buttons, headers, active states.	#2E6B3E
Secondary — Orange	Accent color. Call-to-action buttons, highlights, alerts/notifications, active tab indicators.	#E07A1F
Neutral — Light backgrounds	Card backgrounds, table stripes, section backgrounds.	#F5F5F0 / #E6F0E8
Text	Body copy and headings.	#222222 (body) / #1F4A2C (headings)

Note to Developer

- Confirm exact hex codes with the DietHaven brand sheet before final build — the values above are approximations for design purposes.
- Green should dominate as the primary brand color; orange is used sparingly as an accent to draw attention to actions (e.g., "Save Assessment", "Send to Patient", alerts).
- Maintain WCAG AA contrast ratios, especially orange text/icons on light backgrounds.

3. System Overview

DietHaven Consult is a two-sided clinical application connecting registered dietitians with their patients. It supports the full Nutrition Care Process digitally, with AI features assisting (not replacing) the dietitian's clinical judgment at every stage.

Core User Types

- Super Admin — DietHaven staff who manage the platform, dietitian onboarding, and system configuration.
- Dietitian — Registered nutrition professional who manages patients and delivers the Nutrition Care Process.
- Patient — End user who receives care plans, logs food intake, and tracks progress.

Recommended Platform Approach

- Dietitian-facing app: Web dashboard (desktop-first, responsive) — assessments, care planning, and reporting are data-entry heavy and benefit from larger screens.
- Patient-facing app: Mobile app (Android/iOS) — daily food logging, weight check-ins, and viewing care plans should be optimized for phone use.
- Shared backend: Single API and database serving both apps, ensuring real-time sync between dietitian and patient views.

4. Module 1: DietHaven Consult — Feature Breakdown

4.1 User Management

Handles onboarding, authentication, and access control for all user types.

- Dietitian registration with license/credential verification fields (e.g., professional body registration number).
- Patient registration — self-service sign-up or invited by a dietitian via email/SMS/link.
- Role-Based Access Control (RBAC): Admin, Dietitian, and Patient roles with distinct permissions (see table below).
- Secure authentication: email/password with option for OTP or biometric login on mobile.
- Encrypted storage of all patient records, with audit logging of who accessed or modified a record and when.
- Patient-dietitian linkage: a patient can be linked to one primary dietitian; a dietitian can manage many patients.

Role Permission Matrix

Capability	Admin	Dietitian	Patient
Manage platform settings & dietitian accounts	Yes	No	No
Create/edit own profile	Yes	Yes	Yes
Register and manage own patients	No	Yes	No
View own records only	No	No	Yes
Perform nutrition assessment on a patient	No	Yes	No

Capability	Admin	Dietitian	Patient
View AI diagnosis/intervention suggestions	No	Yes	No
Log daily food intake / weight	No	No	Yes
View own care plan & progress reports	No	Yes (own patients)	Yes (self)

4.2 Nutrition Care Process (NCP)

The NCP module should follow the four standard IDNT stages in sequence, with each patient encounter producing a linked record: Assessment → Diagnosis → Intervention → Monitoring & Evaluation. Each stage's data should be timestamped and versioned so a full care history is preserved.

4.2.1 Nutrition Assessment

The dietitian collects data across six domains. Suggested fields per domain:

Domain	Suggested Data Points
Patient (Client) History	Personal/social history, medical & health history, family history, treatment/medication history
Anthropometric	Height, weight, BMI (auto-calculated), weight history/change, waist circumference, growth pattern (pediatric)
Biochemical / Laboratory	Lab values (e.g., fasting blood glucose, HbA1c, lipid profile, renal/liver panel), test date, reference ranges with flags for abnormal results
Nutrition-Focused Physical (Clinical)	Physical appearance, muscle/fat wasting, swallowing function, oral health, vital signs
Dietary / Food & Nutrition History	Usual intake (24-hr recall or food frequency), meal pattern, food preferences/aversions, allergies, cultural/religious food practices, supplement use
Environmental / Lifestyle	Living situation, food access/security, physical activity level, occupation, socioeconomic factors affecting food choices

Developer Note

- Build assessment forms as structured, stage-gated forms (not one long form) so dietitians can save progress per domain.
- Auto-calculate derived values (BMI, weight change %) rather than requiring manual entry.
- Flag abnormal lab values automatically against standard reference ranges to support the diagnosis step.

4.2.2 Nutrition Diagnosis (AI-assisted)

Nutrition diagnosis should follow the IDNT PES statement structure: Problem (diagnostic label) linked to an

Etiology (root cause) evidenced by Signs/Symptoms gathered during assessment.

- Diagnosis library organized by the three IDNT domains: Intake, Clinical, and Behavioral-Environmental.
- AI suggestion engine analyzes assessment data (esp. flagged labs, dietary intake gaps, and anthropometric trends) and proposes one or more candidate PES statements, ranked by likelihood/relevance.
- Dietitian retains final authority: the system must let the dietitian accept, edit, or reject any AI-suggested diagnosis before it is saved to the patient record.
- Every accepted diagnosis should store which fields of the assessment served as its 'signs/symptoms' evidence, for traceability.

Example PES Statement Structure

- Problem: e.g., "Excessive carbohydrate intake"
- Etiology (related to): e.g., "limited nutrition-related knowledge"
- Signs/Symptoms (as evidenced by): e.g., "24-hour recall showing 65% of energy from refined carbohydrates and HbA1c of 8.2%"
- Full sentence format: "[Problem] related to [Etiology] as evidenced by [Signs/Symptoms]."

4.2.3 Nutrition Intervention

- Personalized nutrition care plan builder — links directly to the accepted diagnosis (each intervention should map to a specific PES problem/etiology).
- Meal planning tool with drag-and-drop or template-based plan builder, calorie/macro targets, and meal timing.
- Nigerian Food Exchange List integration — food database organized into exchange groups (e.g., starches, legumes, vegetables, fruits, fats, proteins common to Nigerian diets) with portion/exchange equivalents, used to build culturally appropriate meal plans.
- Nutrition prescription generator — produces a formal, printable/shareable prescription (energy/macronutrient targets, restrictions, supplement recommendations).
- AI-assisted intervention recommendations — suggests intervention strategies and sample meal plans based on the diagnosis and patient preferences/restrictions, for dietitian review and editing.

4.2.4 Monitoring & Evaluation

- Weight tracking with trend charts over time.
- BMI tracking (auto-calculated alongside weight/height updates).
- Laboratory marker monitoring — trend view of key labs over multiple visits, compared against target ranges.
- Progress reports — auto-generated summary combining anthropometric, biochemical, and dietary adherence trends, exportable as PDF.
- Follow-up documentation — structured notes for each follow-up visit, referencing prior diagnosis/intervention and recording outcome/status (resolved, improved, unchanged, worsened).

4.2.5 Daily Food Monitoring

- Patient-facing food diary — log meals/snacks with photo, description, or search from the food exchange database.
- Adherence tracking against the prescribed meal plan (e.g., % of plan followed per day/week).
- Simple patient-side feedback (hunger/fullness, symptoms) optional per entry.
- Dietitian dashboard view of patient logging activity and adherence trends between visits.

5. AI Features & Clinical Decision Support

AI features are designed as decision support, not autonomous clinical decision-making. The system should always position AI output as a suggestion requiring dietitian review, editing, and sign-off.

AI Feature	Function	Human-in-the-Loop Control
AI Nutrition Diagnosis Suggestions	Analyzes assessment data to propose ranked PES statements	Dietitian must accept/edit/reject before it's saved to the record
AI-Assisted Intervention Recommendations	Suggests intervention strategies tied to the accepted diagnosis	Dietitian edits/approves before it becomes the active care plan
AI-Powered Meal Planning Support	Generates draft meal plans using the Nigerian Food Exchange List, patient preferences, and calorie/macro targets	Dietitian can regenerate, swap items, or manually override any meal
Clinical Decision Support	Surfaces relevant alerts (e.g., abnormal lab trend, unsafe combination) during assessment/intervention	Advisory only — displayed as flags/prompts, never blocks dietitian action

Guardrails for the Development Team

- Log every AI suggestion and the dietitian's action on it (accepted as-is, edited, rejected) — this creates an audit trail and a dataset for improving suggestion quality over time.
- Never auto-save AI output directly into a patient's official record without explicit dietitian confirmation.
- Clearly label AI-generated content in the UI (e.g., an 'AI Suggested' badge) so it's visually distinct from dietitian-entered data.

6. Data & Security Requirements

- All patient health data encrypted at rest and in transit (TLS for transport, AES-256 or equivalent at rest).
- Compliance with the Nigeria Data Protection Act (NDPA) 2023 — informed consent capture at registration, data access/export rights, and a data retention/deletion policy.
- Role-based access control enforced at the API level, not just the UI, so a patient account can never query another patient's data.
- Full audit logs for record creation, edits, and views (who, what, when).
- Regular automated backups with a documented recovery process.

7. Suggested Core Data Entities

A starting point for the database schema — the development team should expand this during technical design:

Entity	Key Fields
User	id, role, name, email, phone, password_hash, status, created_at
DietitianProfile	user_id, license_number, specialty, facility
PatientProfile	user_id, dietitian_id, date_of_birth, sex, contact, consent_status
Assessment	id, patient_id, dietitian_id, date, domain data (history, anthropometrics, labs, clinical, dietary, environmental)
Diagnosis	id, patient_id, assessment_id, PES_problem, PES_etiology, PES_signs_symptoms, domain, ai_generated (bool), status
Intervention	id, patient_id, diagnosis_id, care_plan_details, meal_plan_id, prescription_details, ai_generated (bool)
MealPlan	id, patient_id, meals[], exchange_list_refs[], calorie_target, macro_targets
FoodExchangeItem	id, food_name, exchange_group, portion_size, macronutrients
MonitoringEntry	id, patient_id, date, weight, bmi, lab_values[], notes
FoodLog	id, patient_id, date, meal_type, food_items[], photo_url, adherence_flag
AuditLog	id, user_id, action, entity_type, entity_id, timestamp

8. Suggested Build Phases

1. Phase 1 — Foundations: User management, RBAC, authentication, patient-dietitian linkage.
2. Phase 2 — Core NCP: Assessment forms, diagnosis library (manual entry first), intervention/care plan builder, food exchange list database.
3. Phase 3 — Monitoring: Weight/BMI/lab tracking, progress reports, follow-up documentation, patient food diary.
4. Phase 4 — AI Layer: AI diagnosis suggestions, AI intervention recommendations, AI meal planning, clinical decision support alerts.
5. Phase 5 — Polish & Compliance: Security hardening, NDPA compliance review, audit logging, performance testing, UAT with pilot dietitians.

9. Open Items for Discussion with the Development Team

- Confirm mobile vs. web priority for the first release (recommend: Dietitian web dashboard first, patient app second, or both in parallel if resources allow).
- Confirm which AI model/provider will power the suggestion engine, and data-handling implications of sending patient data to a third-party AI API.
- Confirm the full Nigerian Food Exchange List dataset source so it can be structured into the database ahead of intervention module development.

- Confirm subscription/billing model if dietitians or patients will pay for access, as this affects account tiers in User Management.